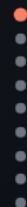




Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

arxiv.org/abs/2504.04715



The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Commercial LLM API providers charge for specific models, but the black-box nature of APIs means users have no guarantee the advertised model is actually served. Providers face strong economic incentives to covertly substitute cheaper alternatives.



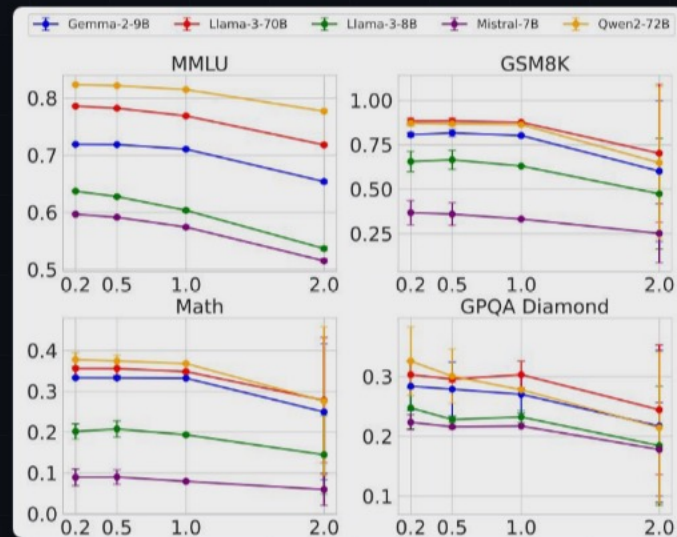
FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

For most models, FP8 scores fall within the error bars of the original, especially on MMLU and GSM8K. The small performance differences make detection extremely challenging.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19



Higher Temperature Amplifies Performance Gaps but Increases Noise



At temperature 2.0, performance degrades across the board and the gap between original and substitute narrows or becomes noisy, complicating statistical detection.

Log Probability Fingerprinting Falls Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The same Llama-3-8B model produces different log probabilities across transformers versions, vLLM releases, and GPU hardware (A100 vs H100). An auditor cannot distinguish legitimate infrastructure differences from actual substitution.



Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Providers can mix substitute responses with probability p . As p approaches 0, the mixed distribution converges to the specified model's distribution.

$$P_{\text{mixed}} = (1 - p) \cdot P_{\text{spec}} + p \cdot P_{\text{substitute}}$$

As $p \rightarrow 0$, the mixed distribution becomes statistically indistinguishable from P_{spec} requiring exponentially more queries for detection.



Software-Only Auditing Methods Are Fundamentally Unreliable

Three classes of detection methods fail under realistic deployment conditions. Each suffers from a distinct, irrecoverable weakness.

Method Type	Weakness	Practical Limitation
Text-based statistical tests (e.g., MMD)	Require impractically large query budgets to achieve statistical power against subtle substitutions	Fail against randomized routing and small-distribution shifts; cost scales super-linearly
Log-probability fingerprinting	Defeated by production-level inference nondeterminism	Same model yields different log-probs across software versions (transformers, vLLM) and GPU hardware (A100 vs H100)
Adversarial countermeasures	Active evasion by the provider undermines all passive detection	Benchmark evasion and randomized routing make any software-only approach query-prohibitive



Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEEs are hardware-secured enclaves that provide cryptographic attestation of the code and data being executed. Unlike software tests, they offer a fundamentally different security guarantee.

- **Provable:** Cryptographic attestation that specific model weights are loaded
- **Efficient:** Only modest performance overhead vs. ZKPs, which are computationally prohibitive for large models
- **Robust:** Immune to randomized substitution and benchmark evasion

NVIDIA Confidential Computing on H100 GPUs is a practical deployment path available today. The provider can prove that the correct inference code is running and that the advertised model weights are in use — without revealing proprietary weights to the auditor.



Key Takeaways: Close the Verification Gap with Hardware Attestation

- 01 Model substitution is economically motivated and practically undetectable via software-only methods.
- 02 FP8 quantization barely affects benchmark scores, making it an attractive but nearly invisible substitution vector.
- 03 Inference nondeterminism across frameworks and hardware defeats log-probability fingerprinting.
- 04 Randomized substitution and benchmark evasion make statistical detection query-prohibitive.
- 05 Trusted Execution Environments are the only currently viable path to verifiable, provable LLM API integrity.

Adopt hardware attestation standards for LLM API transparency.

